

Essential Summary

A Modified Perturbation Method for Global Dynamic Analysis of Generalized Mixed Rayleigh–Liénard Oscillator with Cubic and Quintic Nonlinearities

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1 Background and Motivation

Deriving an analytical relationship between system parameters and the amplitude of limit cycles is a meaningful yet challenging task. When the restoring force or nonlinear damping is complicated, many existing analytical methods **cannot execute their procedures symbolically** — they require all system parameters to be assigned numerically upfront, yielding only isolated snapshots in parameter space rather than a continuous global picture.

This work proposes a **modified generalized harmonic function perturbation method** and applies it to a generalized mixed Rayleigh–Liénard oscillator with both cubic and quintic nonlinearities. This paper is historically significant as the **first publication** introducing the composite Simpson integration modification to the GHFP framework — it predates the IJNLM 2024 paper (which was the second application with a more complex oscillator). The proposed method enables the analytical derivation of amplitude–parameter relationships, allowing quantitative analysis of the existence, stability, number, position, and amplitude of each limit cycle.

2 System Model

The generalized mixed Rayleigh–Liénard oscillator with cubic and quintic nonlinearities:

$$\ddot{x} + g(x) = \varepsilon(\mu + \mu_2 x^2 + \mu_4 x^4 + \mu_{1r} \dot{x}^2 + \mu_{2r} x \dot{x}^2 + \mu_{3r} x^3 \dot{x}^2) \dot{x} \quad (1)$$

where $g(x)$ is the restoring force, μ is the control parameter, and $\mu_2, \mu_4, \mu_{1r}, \mu_{2r}, \mu_{3r}$ are damping coefficients combining Rayleigh-type ($\dot{x}^2$ terms) and Liénard-type (x^2, x^4 terms) damping.

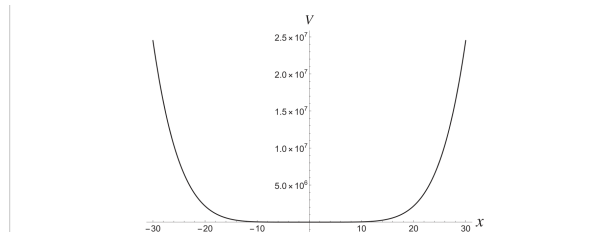


Figure 1: **Figure 1.** Potential portrait of the oscillator under single-well conditions. The restoring force determines the qualitative structure within which limit cycles evolve.

3 The Modified GHFP Method

3.1 Key Innovation

The classical GHFP method fails for complicated oscillators because the integrals involving non-linear terms cannot be evaluated analytically. The **composite Simpson quadrature formula** is introduced to discretize these integrals:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}} f(x_i) + f(x_n) \right] \quad (2)$$

By dividing the integration interval into eight equal subintervals, the Fourier coefficients become **explicit algebraic functions** of the system parameters. The entire perturbation procedure can now be executed purely symbolically.

3.2 Procedure

1. Nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$
2. Solution expansion $x = a \cos^2 \varphi + b$
3. Perturbation expansion $a = a_0 + \varepsilon a_1 + \dots$
4. Composite Simpson integration to compute Fourier coefficients

4 Key Equations

4.1 Amplitude–Parameter Relation

$$\mu = -\frac{1}{a_0^2(2p_0 - p_4)} \left(E_2\mu_2 + E_4\mu_4 + E_{1r}\mu_{1r} + E_{2r}\mu_{2r} + E_{3r}\mu_{3r} \right) \quad (3)$$

where p_{2i} are Fourier coefficients computed via composite Simpson integration and E_i are functions of a_0 , b , and p_{2i} . Given system parameters, the limit cycle amplitude is obtained analytically.

4.2 Stability Criterion

$$h_0 = \frac{1}{\pi} \int_0^\pi \left[\mu + \sum \mu_i (a_0 \cos^2 \varphi + b)^i + \sum \mu_{jr} (a_0 \cos^2 \varphi + b)^j (\dot{x}_0)^2 \right] d\varphi \quad (4)$$

$h_0 > 0$: unstable; $h_0 = 0$: semi-stable; $h_0 < 0$: stable.

4.3 Homoclinic and Heteroclinic Bifurcation

Setting the limit cycle amplitude equal to the saddle point coordinate in Eq. (3) yields the critical μ^{hom} for homoclinic bifurcation. Similarly, heteroclinic bifurcation thresholds are obtained when the amplitude equals the coordinates of two connected saddle points.

5 Example 1: Single-Well Potential (Section 4.1)

The oscillator (53) with single-well potential is analyzed first. The μ – A curve reveals the complete limit cycle lifecycle:

Key observations:

- Below a critical μ , **no limit cycle exists**

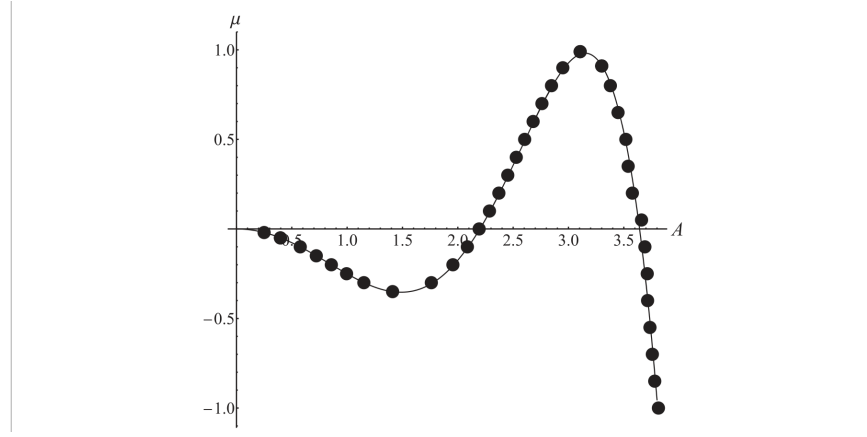


Figure 2: **Figure 4.** Comparison of μ - A curves for the single-well oscillator ($\varepsilon = 0.01$). Solid lines: present method. Dotted lines: Runge-Kutta reference.

- At the critical value, a **semi-stable** limit cycle emerges
- As μ increases, it splits into stable and unstable cycles
- The unstable cycle eventually **collapses** to the equilibrium; the stable cycle persists
- Up to **three coexisting limit cycles** are possible under certain parameter configurations (Figure 5 in the original paper)

6 Example 2: Triple-Well Potential (Section 4.2)

The oscillator (66) with triple-well potential presents a more complex landscape.

The triple-well case demonstrates that the method can handle:

- Multiple coexisting limit cycles within different potential wells
- Both homoclinic and heteroclinic bifurcation thresholds
- Asymmetric limit cycles (not all enclosed by a single homoclinic orbit)

7 Key Findings

1. **First composite Simpson GHFP variant:** This is the first paper to introduce the composite Simpson quadrature formula into the GHFP framework, enabling purely symbolic execution for complicated oscillators. It directly predates and enables the IJNLM 2024 work.
2. **Complete lifecycle prediction:** The analytical μ - A relationship reveals the entire lifecycle of each limit cycle — from generation (semi-stable bifurcation) through evolution (stable and unstable branches) to termination.
3. **Up to three coexisting limit cycles:** The method identifies parameter regions where three limit cycles coexist simultaneously — a complex dynamical scenario that purely numerical methods struggle to capture completely.
4. **Homoclinic/heteroclinic bifurcation prediction:** Both types of global bifurcation are quantitatively analyzed. The analytical expressions serve as precursors to chaos, with practical implications for chaos control.
5. **Dual applications:** The framework supports both (i) **limit cycle regulation** (inertial impact shaker design, population dynamics) and (ii) **chaos prediction** via homoclinic/heteroclinic bifurcation thresholds (encryption, equipment protection).

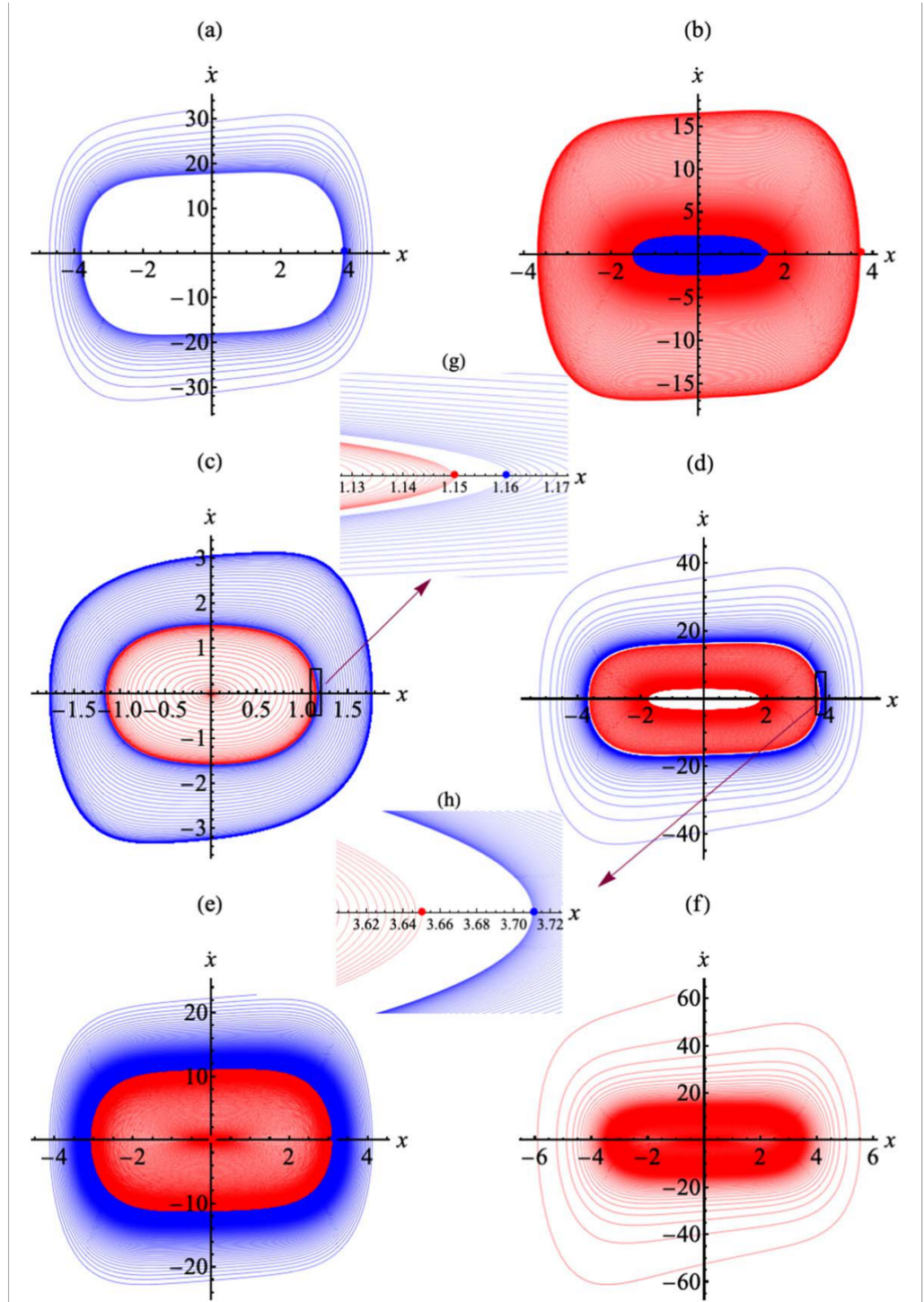


Figure 3: **Figure 3.** Phase portraits of the single-well oscillator with different μ and A values ($\varepsilon = 0.01$). The colorized dots denote the initial point of each trajectory. The evolution from no limit cycle to a single stable cycle is clearly visible.

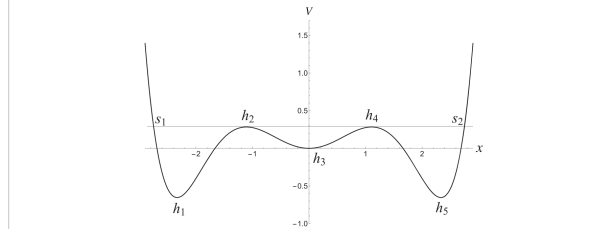


Figure 4: **Figure 6.** Potential portrait of the triple-well oscillator. The multiple wells create a rich bifurcation structure where both homoclinic and heteroclinic orbits can coexist.

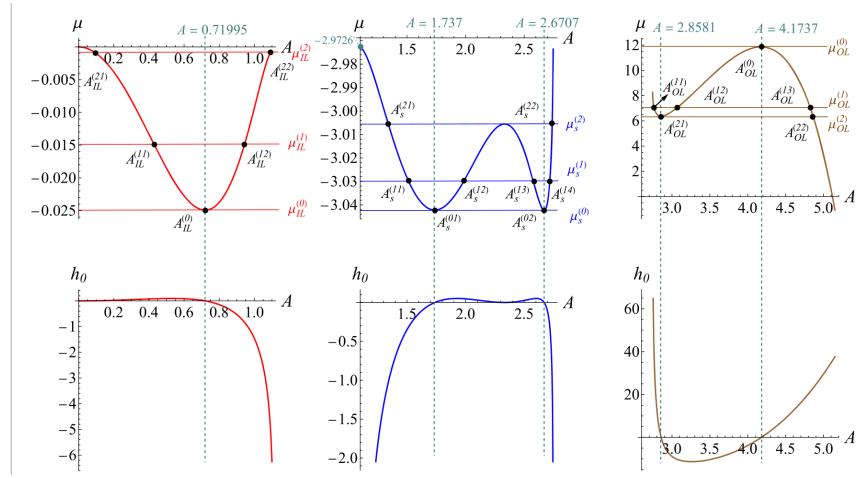


Figure 5: **Figure 7.** The μ - A and h_0 - A curves for the triple-well oscillator. The evolution and stability of limit cycles are quantitatively captured across the entire parameter range.

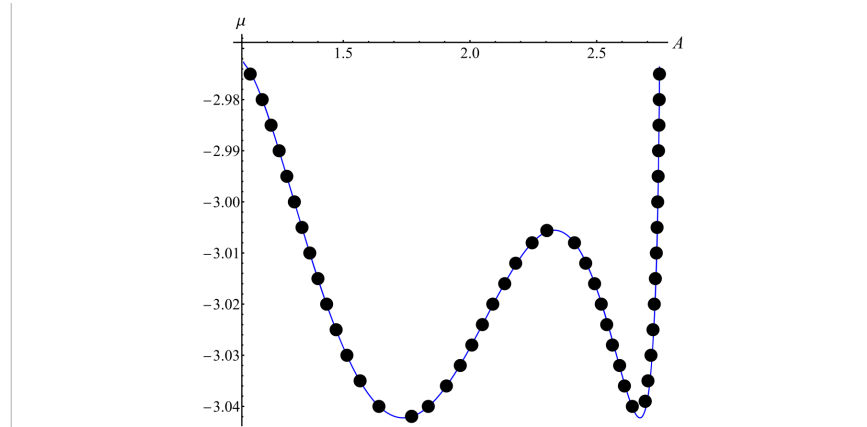


Figure 6: **Figure 9.** Comparison of μ - A curves for limit cycles between saddle points in the triple-well oscillator ($\varepsilon = 0.1$). Solid lines: present method. Dotted lines: Runge-Kutta. The method accurately predicts bifurcation thresholds.

8 Significance and Future Work

This paper represents the **first publication** of the MGHFP methodology. The proposed method compensates for the insufficiency of existing perturbation-based approaches and can be adopted for analyzing many other strongly nonlinear oscillators, including Duffing-harmonic oscillators (rational restoring force) and Helmholtz–Duffing oscillators (asymmetric potential) — topics addressed in subsequent work.

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